

Diagnosis of Penetrating Cervical Esophageal Injuries

John A. Weigelt, MD, FACS, Erwin R. Thal, MD, FACS, William H. Snyder, III, MD, FACS, Richard E. Fry, MD, Donald E. Meier, MD, FACS, and William J. Kilman, MD, Dallas, Texas

Management options for patients with penetrating neck trauma include mandatory exploration and selective observation [1-4]. Disagreement over management plans primarily concerns the patient with minimal or no findings of airway, gastrointestinal, or vascular injuries. Airway injuries are fairly obvious when clinically important. Arteriography accurately excludes arterial injuries [5,6]. Esophageal injury remains a diagnostic dilemma as the accuracy of tests to identify this injury is unknown. The risk of missing an esophageal injury is often used to justify mandatory exploration of all injuries penetrating the platysma [1,7].

This study was designed to prospectively evaluate the accuracy of barium swallow and endoscopy in detecting esophageal injury after penetrating cervical trauma.

Material and Methods

One hundred eighteen patients with penetrating neck wounds in zones II and III were evaluated at Parkland Memorial Hospital, Dallas, Texas, from July 1981 to December 1984. There were 96 male patients and 22 female patients with an average age of 28 years. Seventy-seven patients (65 percent) had been stabbed and 41 (35 percent) had been shot. All patients had stable vital signs without altered sensorium and a cervical wound in proximity to the esophagus. Patients with continued bleeding consistent with major vascular injuries in the neck or airway difficulty were taken directly to the operating room and excluded from the study.

Patients were enrolled in the study after initial resuscitation and assessment were complete. Informed consent was obtained as outlined by the Institutional Review Board. Cervical biplane angiography, followed by barium swallow and fluoroscopy with cineradiography in two projections were performed. All patients were taken to the operating room for exploration. After induction of anesthesia and endotracheal intubation, fiberoptic and rigid endoscopy were performed by an endoscopist who had no knowledge of the barium swallow results. The esophagus was defined as starting below the cricopharyngeal mus-

cles. Any finding during endoscopy that the endoscopist could not explain as iatrogenic trauma was considered to indicate an injury. After all studies had been completed, a neck exploration was performed.

The results were reviewed to assess the accuracy of endoscopy and barium swallow to diagnose an esophageal injury. Each patient's chart was reviewed at discharge and at least 30 days after discharge. The sensitivity, specificity, and accuracy was calculated for each diagnostic test. Sensitivity was expressed as the number of true-positive results divided by the number of true-positive results plus the number of false-negative results. Specificity was expressed as the number of true-negative results divided by the number of true-negative results plus the number of false-positive results. Accuracy was expressed as the number of true-positive and true-negative results divided by the total number of patients.

Results

There were no organ injuries at cervical exploration in 79 patients (67 percent). The remaining 39 patients (33 percent) had 40 injuries, including 10 esophageal, 2 pharyngeal, 4 laryngeal, 4 tracheal, 8 arterial, 6 venous, 3 peripheral nerve, and 3 spinal cord injuries.

Symptoms of esophageal injury included dysphagia, hematemesis, hoarseness, and odynophagia. Signs included subcutaneous air or crepitus, retropharyngeal air, retropharyngeal edema, hematoma, deviated trachea, and pneumomediastinum. Seven of the 10 patients (70 percent) with an esophageal injury had symptoms or signs of injury. All patients with an esophageal injury from a gunshot wound had clinical findings of an esophageal injury, whereas 50 percent of the patients with stab wounds injuring the esophagus had no such findings. Twenty-five percent of the patients without esophageal injuries had symptoms and 26 percent had signs suggesting an injury.

One hundred twelve patients (95 percent) had barium swallows, 106 (90 percent) had flexible endoscopies, and 116 (98 percent) had rigid endoscopies. A total of 93 patients (79 percent) had all three preoperative studies. Among patients with esophageal injuries, the results of barium swallow were positive in eight and negative in one. Results of flexible endoscopy were positive in three patients and negative in five. Results of rigid endoscopy were

From the Departments of Surgery and Radiology, Southwestern Medical School, The University of Texas Health Science Center, Dallas, Texas.

Requests for reprints should be addressed to John A. Weigelt, MD, FACS, University of Texas Health Science Center, Department of Surgery, 5323 Harry Hines Boulevard, Dallas, Texas 75235.

Presented at the 39th Annual Meeting of the Southwestern Surgical Congress, Coronado, California, April 26-29, 1987.